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Selective distribution of pharmacy item data from pharmacy item tracking system

Abstract

Methods and systems relate to providing varying levels of detail about a pharmacy item to a patient information system based on the capabilities of the patient information system. An intercept device can receive an identifier associated with a pharmacy item based on a scanner reading a computer-readable code. The intercept device can recognize or determine the storage capabilities or limitations of the patient information system, for example by determining various medication data fields that the patient information system uses to store pharmacy item information. The intercept device can use the machine readable code to obtain pharmacy item information that is compatible with the patient information system based on those storage capabilities or limitations and can distribute that pharmacy item information to the patient information system.

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Primary Examiner: Collins; Michael

Attorney, Agent or Firm: Knobbe, Martens, Olson & Bear, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/151,096, entitled “SELECTIVE DISTRIBUTION OF PHARMACY ITEM DATA FROM PHARMACY ITEM TRACKING SYSTEM,” filed on Jan. 6, 2023, which is a continuation of U.S. patent application Ser. No. 16/985,594, entitled “SELECTIVE DISTRIBUTION OF PHARMACY ITEM DATA FROM PHARMACY ITEM TRACKING SYSTEM,” filed on Aug. 5, 2020, which claims priority benefit to U.S. Provisional Application No. 62/883,351, entitled “SELECTIVE DISTRIBUTION OF PHARMACY ITEM DATA FROM PHARMACY ITEM TRACKING SYSTEM,” filed Aug. 6, 2019, each of which is hereby incorporated herein by reference in its entirety. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are incorporated by reference under 37 CFR 1.57 and made a part of this specification.

BACKGROUND

(1) Tracking the location, status, and/or other information regarding individual items is often beneficial. Such tracking can often be complex, as in some applications the number of individual items to be tracked can be thousands, millions, or more. As one example, in a healthcare facility, pharmacy items, such as, medications, drugs, diluents, medical and surgical supplies, gauze, scissors, needles, labels, baggies, bandages, packaging, vials, syringes, and/or other medical items for which a pharmacy is responsible, that are distributed from the pharmacy should be managed closely to conform to regulatory guidelines regarding use and waste, to avoid misadministration and diversion, and to ensure appropriate inventory management and patient billing. As another example, shipping and receiving operations process many individual items per day, routing them through various preparation, packaging, routing, shipping, and delivery stages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram illustrating an environment for tracking items and/or selectively communicating medication data field values to a patient information system.
- (2) FIG. 2 illustrates is a data flow diagram illustrative of an example process for selectively distributing medication data field values to a patient information system.
- (3) FIG. 3 is a flow diagram illustrating an embodiment of a routine for selectively communicating medication data field values based on the medication data fields that a patient information system uses to store data.

DETAILED DESCRIPTION

- (4) Systems and methods are described herein for tracking individual items and/or selectively distributing data associated with individual items. Although various aspects of the disclosure will be described with regard to examples and embodiments, one skilled in the art will appreciate that the disclosed embodiments and examples should not be construed as limiting.
- (5) In the present disclosure, reference is made to items. As used herein, the term “item” is used broadly to define any article, person, process, or procedure that may be tracked using a tracking system. In some examples described herein, an item can be a pharmacy item.
- (6) The term “pharmacy item” is used broadly to refer to any item that is distributed from a pharmacy or

used in the medical treatment of a patient in a healthcare facility. By way of non-limiting example, a pharmacy item can be a medication, drug, diluent, or other medical or surgical supply, including gauze, scissors, needles, syringes, labels, baggies, bandages, medical packaging, vials or other medical containers, or any other healthcare item. Throughout this disclosure, examples are described wherein a pharmacy item is a medication, or dose thereof, to be administered to a patient; however, these examples are not intended to be limiting, and a pharmacy item can be a non-medication healthcare item used in the treatment of a patient. However, a pharmacy item is merely a subset of the items contemplated within this application and tracking of other types of items, in other contexts, is also within the scope of this disclosure. For example, an item may refer to any package or product processed in manufacturing operation, shipping and receiving operation, storage operation, etc. Moreover, in some instances, the term item is used broadly to encompass tracking of people or animals. For example, the tracking systems described herein may be used to track a patient or healthcare professional within a healthcare facility. In some instances, item may refer to a process or procedure. For example, in the healthcare context, an item may refer to a particular test prescribed for a particular patient.

(7) In some instances, items can contain or be associated with one or more machine-readable codes or labels, for example, barcodes (e.g., but not limited to, one-dimensional barcodes, two-dimensional barcodes, such as, QR codes, etc.), radio-frequency identification (RFID) tags, etc. For example, individual items can include a code or an RFID tag having machine-readable information such as an item identifier, intended user, or location, for example. A container such as a box, tray, or canister that includes pharmacy items can include a code or an RFID tag having machine-readable information such as a kit identifier, kit type, intended user, or location, for example.

(8) A machine-readable code, label, or tag can include an identifier that corresponds to a particular patient with whom the item(s) is/are associated. The machine-readable code can include identifiers that correspond to product identifiers (e.g., National Drug Codes (NDC) or Universal Product Code (UPC)), lot numbers, and expiration dates. The identifiers can correspond to medications listed in a pharmacy item order, such as, NDC. In some instances, identifiers can correspond to one or more unique identifiers associated with the pharmacy items or doses thereof, which are generated by a tracking system and used to track the pharmacy item. A pharmacist or other user can verify the contents of a pharmacy item for accuracy. The identifiers or other information can be associated with the code or RFID tags in a computer database to allow subsequent identification and processing by RFID or other technology.

(9) As used herein, the term “unique identifier” or similar term, is intended to refer to an identifier, such as an alpha-numeric identifier, that can uniquely identify an item from other items. In some cases, a unique identifier can uniquely identify an item from other items that include the same type of drug or from other items that include the same drug. In some cases, a unique identifier can uniquely identify an item from all other items. For example, an NDC is used to identify different drugs and a UPC is used to identify different products. However, an NDC is unable to identify a particular vial, syringe, bag, or dose, of one type of drug from a different vial, syringe, bag, or dose, of the same kind of drug. Similarly, a UPC is unable to identify one instance of a product from another instance of the same product. The unique identifier, however, can uniquely identify a particular vial, syringe, bag, or dose from all other vials, syringes, bags, or doses of a particular drug and other drugs or pharmacy items, and in some cases, may also be referred to as a universally unique identifier or item-specific unique identifier. Further, the unique identifier can be embedded in a machine-readable label, which may also be referred to herein as a machine-readable unique identifier. In some cases, the unique identifier only uniquely identifies an associated item within the tracking system.

(10) A pharmacy item can be released to the healthcare facility to be administered to the patient. In some instances, the pharmacy item is released to a nursing station or other pharmacy item drop off location within a healthcare facility. From the nursing station or other pharmacy item drop off location, the pharmacy item can be picked up by a nurse, a doctor, or other healthcare professional and delivered and administered to the patient. In some instances, custody of the pharmacy item is handed off between various healthcare professionals multiple times after being received in the healthcare facility but before arriving at the patient. For this reason, it can be difficult to track a pharmacy item throughout the entire process of entry, preparation, and delivery and administration to the patient. The methods and systems described herein can allow for increased accuracy in tracking the pharmacy item in these circumstances.

(11) When the pharmacy item is to be administered to the patient, the administering healthcare professional can scan one or more of the machine-readable labels printed on or associated with the pharmacy item. The healthcare professional can also scan one or more machine-readable codes or labels associated with the patient, for example a patient identifier encoded within a machine-readable label, which can be located on a wristband on the patient. In some instances, by scanning both a machine-readable code or label associated with a pharmacy item and a machine-readable label associated with the particular patient, the system can verify that the pharmacy item can be administered to that particular patient. For example, the system can verify that the item being administered is not expired, past its beyond-use date, or recalled/blacklisted.

(12) Embodiments of the system described herein can be used to track items from preparation, delivery, and/or other processing steps. In some instances, the systems and methods described herein can be used to track the location and/or other status of particular items.

(13) In some instances, at the various stages in the preparation and delivery process, machine-readable unique identifiers can be scanned to indicate particular actions, for example, arrival at a particular location, custody exchange, or administration to a patient. When scanned, a pharmacy item tracking system can update the tracking database entries to include additional information provided by the scan. For example, once the pharmacy item is delivered to a nursing station, a machine-readable unique identifier associated with the pharmacy item can be scanned to indicate that the pharmacy item has reached the nursing station. When a particular nurse, picks up the pharmacy item for delivery to the patient the pharmacy item can be scanned again and the tracking database entry can be updated to include information about which nurse has taken custody of the pharmacy item. When the pharmacy item is delivered to the patient, the machine-readable unique identifier associated with a particular pharmacy item can be scanned to indicate that the particular pharmacy item has been (or is about to be) administered to a patient. When the pharmacy item or a portion thereof is leftover, the machine-readable unique identifier associated with a particular pharmacy item can be scanned to indicate that the particular pharmacy item has been (or is about to be) returned or disposed of.

(14) In some instances, the systems and methods as described herein can be integrated into existing systems without requiring modification of the existing systems. For example, the systems and methods described herein can intercept communication between existing hardware at the facility, parse the communication to extract item information therefrom, obtain unique identifiers associated with the item, and generate and/or update tracking database entries associated with the item, while maintaining the expected communication between existing healthcare facility hardware.

(15) Although the preceding description has outlined an example of a tracking system at a healthcare facility and used to track pharmacy items, it will be appreciated that the tracking system need not be limited to such an application. For example, the tracking systems as described herein can be used within a healthcare facility to track non-pharmacy items. For instance, the tracking systems may be used in a healthcare facility to track wheelchairs, beds, food orders, etc. Moreover, the tracking systems can be used to track patients themselves, or to track a patient's progress through various stages of treatment. Further, the tracking systems may be used to track patient samples, for example, tissue samples, blood samples, etc., as they are processed. The tracking systems described herein also have wide applicability in non-healthcare applications. For example, the tracking systems can be employed in manufacturing operations, shipping and receiving operations, warehouse and storage operations, etc. For example, the tracking system can be used to track individual component parts as they are assembled in manufacturing process, individual packages to be shipped in a distribution network, or the location and movement of individual items within a warehouse.

(16) In some cases, where the items to be tracked are already associated with codes that identify them uniquely, the tracking systems and methods described herein may make use of these existing codes. For example, in a hospital a patient may be assigned a unique patient code during intake. A machine-readable label including the embedded unique patient code may be affixed to the patient, for example, with a wristband. Hospital procedures may require scanning of the machine-readable label at various stages of the patient's treatment, for example, before administering medication to the patient. When the machine-readable label is scanned, information from the scan may be sent to a hospital system for processing, for example, to determine whether the medication about to be administered is recalled, past its beyond use

date, or recalled/blacklisted. The tracking system described herein can intercept this transmission, create and/or modify tracking database entries, for example, by updating an entry related to the patient's location, and then retransmit the information to the existing hospital system. Accordingly, the tracking system is able to provide additional tracking functionality using existing unique codes and machine-readable labels, without disrupting the functionality of the existing hospital systems. By co-opting existing unique codes and machine-readable labels where available, the tracking systems described herein are able to overload those codes or labels with additional tracking information.

(17) Systems and methods will now be described in greater detail, by way of example, in reference to several embodiments shown in the figures.

(18) Intercept Device

(19) As described herein, a pharmacy item can include a machine readable code that can be scanned by the scanning device. In some cases, the scanning device scans the machine readable code of the pharmacy item and communicates machine readable data to a patient information system, such as an electronic medical records (EMR) computer. The patient information system can receive and store the machine readable data. In addition, the patient information system can utilize the machine readable data to lookup additional data in another database. For example, the patient information system may be able to use the database to translate a specific drug associated with the machine readable data (for example, Tylenol) into a generic drug (for example, acetaminophen).

(20) Embodiment disclosed herein provide systems, methods, and apparatuses for implementing an intercept device to enable tracking of pharmacy items and/or selective distribution of data associated with those pharmacy items to the patient information system. The intercept device can recognize or determine the storage capabilities or limitations of the patient information system. For example, the intercept device can determine which categories of data that the patient information system uses to store pharmacy item information. And the intercept device can use the machine readable code to obtain pharmacy item information that is compatible with the patient information system based on those storage capabilities or limitations, and can distribute that pharmacy item information to the patient information system. In this way, the patient information system receives its pharmacy item information from the intercept device, without performing its own data lookup. In addition to reducing the amount of computing resources or computing time utilized by the patient information system, in some cases the intercept device has access to pharmacy item information that would otherwise be unavailable to the patient information system. This can improve the safety of the patient information system, for example, by identifying pharmacy items that are expired or recalled. Thus, the implementation of the disclosed intercept device can advantageously improve the robustness of information of the patient information system. In addition, by identifying the capabilities of the patient information system and providing it with mostly (or, in some cases, only) those categories of information that are compatible with its capabilities, the intercept device streamlines and reduces a likelihood of error during data distribution. Furthermore, in some cases, an intercept device can be integrated into an existing healthcare system, thereby limiting costs associated with a system replacement.

(21) An intercept device can act as an intermediary between a scanning device and a patient information system by receiving an identifier from the scanning device and distributing data to the patient information system. For example, the intercept device can receive an identifier associated with a pharmacy item from a scanning device. The intercept device can use the identifier to gather medication data field values for the particular pharmacy item and then can distribute some or all of those medication data field values to the patient information system. By providing the patient information system with medication data field values (instead of, or in addition to, the identifier received from the scanning device), the intercept device advantageously reduces the processing and/or tasks required by the patient information system, since the patient information system does not need to perform its own data lookup using the identifier.

(22) Furthermore, in some instances, the intercept device can determine which medication data fields the patient information system uses to store medication data field values. For example, the intercept device can query the patient information system to determine which medication data fields that the patient information system uses to store information. As another example, the intercept device can determine the medication data fields, based on the type, manufacturer, version or other information about the patient information system. In some cases, this information may be stored in a lookup table. The intercept device

can then distribute those medication data field values (and, in some cases, only those medication data fields) that the patient information system is configured to store. In this way, the intercept device provides the patient information system with information that the patient information system is configured to store (e.g., based on the capabilities of the patient information system). In some cases, intercept device distributes one or more medication data field values that the patient information system is not configured to store. In some such cases, the patient information system can ignore or regard some or all of those medication data field values, or can create additional data fields for those medication data field values.

(23) Environment Overview

(24) FIG. 1 is a block diagram illustrating an environment **100** for tracking items and/or selectively communicating medication data field values to a patient information system **120**. In the illustrated embodiment, the environment **100** includes a scanning device **110**, a patient information system **120**, an intercept device **130**, and a tracking system **140**. The tracking system **140** includes a tracking database **142**. Fewer or more devices or systems can be used in the environment **100** as desired.

(25) Some or all of the scanning device **110**, the patient information system **120**, the intercept device **130**, or the tracking system **140** can be communicatively coupled, for example, via a wired or wireless network. The network can include a local area network, for example, a private network, or a wide area network, including, for example, the Internet. In certain embodiments, some or all of the scanning device **110**, the patient information system **120**, the intercept device **130**, or the tracking system **140** can form part of a single device and communicate via a computer bus. Regardless of the configuration, each separate device can include one or more processors and a computer-readable medium that stores specific computer-executable instructions for carrying out the operations described herein.

(26) The scanning device **110** can be any device configured to scan machine-readable codes, labels, or tags and retrieve machine-readable data therefrom. For example, the scanning device **110** can be an RFID reader or RFID reading station. As another example, the scanning device **110** can include a barcode reader. For example, the scanning device **110** can include a one-dimensional barcode reader or a two-dimensional barcode reader, such as a QR code reader. The scanning device **110** can include a light source, such as a laser, that emits light and a light detector, such as a photodiode, to detect light reflected from the barcode or other machine-readable code. The scanning device **110** can use the detected light to determine the machine-readable data from the barcode or other machine-readable code. In certain embodiments, the scanning device **110** can include an image sensor, such as a charge-coupled device (CCD) image sensor or a complementary metal-oxide semiconductor (CMOS) image sensor. The image sensor can capture an image of the barcode or other machine-readable code, and the scanning device **110** can parse the captured image to extract the machine-readable data. The scanning device **110** can include an RFID tag reader. For example, the scanning device **110** can include or control an antenna that emits a radio frequency signal, which causes an RFID tag to respond with machine-readable data. The machine-readable data can be used to track an item at various preparation, transport, or other processing stages, or a set of items at various preparation, transport, or other processing stages.

(27) The machine-readable data can include any type of data, such as a unique identifier, generic identifier, National Drug Code (NDC), Universal Product Number (UPC), patient data, drug data, etc. In some instances, the scanning device **110** can include a communication interface whereby the scanning device **110** can communicate the machine-readable data to other devices in the environment **100**, such as, the patient information system **120** or the intercept device **130**. The communication interface may allow the scanning device **110** to communicate wireless or via a wired connection.

(28) In some instances, the scanning device **110** can be associated with a particular person. For example, the scanning device **110** can be associated with a healthcare professional, such as a nurse or doctor. In some instances, the scanning device **110** can be associated with a particular location. For example, the scanning device **110** can be associated with a processing station, a nursing station, or patient's room. In some instances, the scanning device **110** may not be associated with a particular location or person. Rather, machine-readable codes can be associated with locations or people and used to associate particular scans with particular locations or people.

(29) As an example, a scanning device **110** at a nursing station can scan a machine-readable code or label that includes an embedded unique identifier associated with an item. The scanning device **110** can interpret machine-readable data (which can include, for example, the unique identifier) from the machine-

readable code or label. This data can be sent to a tracking system and used to indicate that the item has been received at the nursing station.

(30) As another example, a particular nurse or other healthcare professional can scan a machine-readable code including a unique identifier associated with an item to indicate that the particular nurse or other healthcare professional is taking custody of the item (e.g., to deliver the item to the patient), or to indicate that the particular dose is being administered to the patient, etc.

(31) The patient information system **120** can include a computing device used to enter, generate, receive, and/or otherwise process an item. The patient information system **120** can be implemented as personal computer, including a desktop or laptop, a tablet, a mobile phone, or other portable computing device. The computing device can be implemented as a server, or similar networked processing device, that receives and processes an item entered by a remote user. The patient information system **120** can include a computer-readable medium storing computer executable instructions that when executed, enable a computing device to generate, receive, and/or process the item. In some cases, the patient information system **120** can communicate with a printer to print a label or tag for an item, or print an item order. For example, the patient information system **120** can generate a print file, which is sent to the printer and the printer can print the label, tag, or item order based on the print file.

(32) The patient information system **120** can include a database such as a database for electronic medical records (EMRs). EMRs can be a digital version of the paper charts in the clinician's office, or can include patient data, medical history data, treatment history data, or the like.

(33) The database can store, or be configured to store, medication data field values that correspond to a particular set of medication data fields (sometimes referred to as categories of data). The medication data fields can include, but are not limited to, drug name, dosage, hospital, bin number, expiration dates, NDC, UPC, GS1 barcode, location, patient or recipient information, recall information, use-by date, identifier for healthcare provider that is/was in possession of it, or the like. In some cases, the patient information system **120** stores the same medication data fields for each pharmacy item for which it stores data. For example, for each pharmacy item, the patient information system **120** can store one or more medication data field values for each medication data field it includes. However, in some cases, the patient information system **120** stores the different medication data fields for different pharmacy items.

(34) Each medication data field can include one or more medication data field values. For example, a medication data field can include medication data field values for a plurality of pharmacy items. As an example, a medication data field “drug name” can include medication data field values of “Tylenol,” “Propofol,” “Fentanyl,” etc.

(35) In some cases, the patient information system **120** stores a limited number or particular set of medication data fields. The particular set of medication data fields that the patient information system **120** stores can be based on any number of factors including, but not limited to, the capabilities of the patient information system **120**, its manufacturer, version, or the data available to the patient information system **120**. For example, capabilities of the patient information system **120** may limit the categories, amount, types, etc. of information that the patient information system **120** can receive, store, and/or process. As a non-limiting example, in some cases, capabilities of the patient information system **120** may limit it to receiving, storing, or processing NDC numbers. In some such cases, communicating medication data field values that do not correspond to the medication data field of “NDC number” may be ignored, discarded, cause an error, etc.

(36) The intercept device **130** can form part of the patient information system **120** or the scanning device **110**, or can be located along the communication pathway between the patient information system **120** and the scanning device **110**. The intercept device **130** can be configured with computer-executable instructions to receive or intercept the machine-readable data extracted from the machine-readable code or label, for example before any data is provided to the patient information system **120**. The intercept device **130** can be configured with computer-executable instructions to parse the machine-readable data.

(37) The intercept device **130** can identify which medication data fields that the patient information system **120** uses to store medication data field values. This can be accomplished using any of various techniques. For example, the intercept device **130** can query the patient information system **120** to request an indication of the medication data fields it uses. In some instances, the intercept device **130** can query the patient information system **120** by providing the patient information system **120** with the unique

identifier. In response to the query, or in some cases without receiving the query, the patient information system **120** can communicate an indication of the categories it uses.

(38) As another example, the intercept device **130** may be able to consult a lookup table to identify which medication data fields that the patient information system **120** uses to store medication data field values. For example, the intercept device **130** may have previously communicated with the patient information system **120**, identified the categories, and stored an indication of the categories in a lookup table or other storage.

(39) The intercept device **130** can identify the medication data fields that the patient information system **120** uses, or is configured use, to store data at any time, such as before, after, or while concurrently receiving the machine-readable data from the scanning device **110**. For example, the intercept device **130** can be configured to determine the medication data fields or medication data field values accepted by the patient information system **120** as part of a set up procedure and/or prior to the scanning device **110** performing a scan.

(40) The intercept device **130** can communicate a request to the tracking system **140** for the medication data field values that correspond to the medication data fields that the intercept device identified as being used by the patient information system **120**. The request to the tracking system **140** can include an identifier of the medication data fields and a pharmacy item identifier. For example, the request can include some or all of the machine-readable data, such as a unique identifier, that the intercept device **130** receives from the scanning device **110**. The intercept device **130** can relay the medication data field values received from the tracking system **140** to the patient information system **120**. In addition or alternatively, the intercept device **130** can instruct the tracking system **140** to communicate the medication data field values to the patient information system **120**.

(41) As an example, if the intercept device **130** determines that the patient information system **120** uses only three medication data fields (for example, NDC, drug name, and expiration date), then the intercept device **130** can determine to query the tracking system **140** for medication data field values that correspond to at least those (or, in some cases, only those) three medication data fields. As a corollary, if the intercept device **130** determines that the patient information system **120** does not include a medication data field corresponding to expiration date, then the intercept device **130** can determine not to query the tracking system **140** for medication data field values that correspond to the expiration date. In this way, the intercept device **130** can dynamically determine which medication data fields that the patient information system **120** uses, thereby identifying which data to communicate to the patient information system **120** or which information to refrain from communicating to the patient information system **120**.

(42) The tracking system **140** can include information linking individual items with corresponding generic identifiers. For example, the tracking system **140** may include entries that relate a particular medication to its corresponding NDC or the tracking system **140** may include a tracking database **142** or NDC database. As another example, the tracking system **140** can include entries that relate the trade items with corresponding UPCs, GS1 barcodes, etc. The tracking system **140**, the intercept device **130**, or the patient information system **120** can be communicatively coupled to a generic identifier database, such as an NDC database.

(43) In some instances, the tracking system **140** is separate from the intercept device **130**. In some instances, the intercept device **130**, and the tracking system **140** can be integrated into a single device. In some instances, the tracking system **140** is remotely located from the intercept device **130** or the other components in the environment **100**. In some instances, the tracking system **140** is locally located relative to the intercept device **130** or the other components in the environment **100**.

(44) The tracking system **140** can use the request received from the intercept device **130** to update entries in the tracking database **142** or verify information related to the item. For example, the tracking system **140** can identify the unique identifier from the request and can track an item using the unique identifier.

(45) The tracking system **140** can use the request received from the intercept device **130** to select medication data field values that both correspond to the particular pharmacy item associated with the unique identifier and correspond to the medication data fields identified as being used by the patient information system **120**. For example, based on the request from the intercept device **130**, the tracking system **140** can identify data corresponding to one or more entries in the tracking database **142**.

(46) The tracking system **140** can be configured to perform verifications. For example, where the item

includes a medication to be administered to a patient, the tracking system **140** may verify that the particular medication is safe to be administered to a patient. For example, the tracking system **140** may verify pharmacy items by checking for expired or soon-to-be expired medications (for example, based on manufacturer's expiration date or other beyond-use dates), checking for recalled (or otherwise blacklisted) medications, etc. In some instances, when the tracking system **140** indicates that the particular medication is not safe to be administered to the patient, (for example, if the patient is allergic to the medication), the tracking system **140** can provide a notification (for example, an alarm) that the medication should not be administered.

(47) The scanning device **110**, intercept device **130**, and tracking system **140** can be used together, to track items at various preparation, transport, or other processing stages. Additionally, in some instances, the intercept device **130** and the tracking system **140** can be integrated into existing systems. For example, where a facility already includes scanning devices **110** which are configured to communicate with the patient information systems **120**, an intercept device **130** and tracking system **140** as described herein can be integrated into the existing system and provide additional functionality.

(48) In some cases, the intercept device **130** and/or tracking system **140** are integrated into the patient information system **120**. For instance, the intercept device **130** can include software that is running on the patient information system **120**, for example, along with EMR software, and can intercept some or all keystrokes of the patient information system **120** and analyzes them. The scanning device **110** can scan the machine readable code of the pharmacy item and can communicate the machine readable data as keystrokes, for example, since barcode readers can imitate a keyboard.

(49) The scanning device **110** can imitate or correspond to a keyboard and the intercept device **130** can intercept or receive the “keystrokes” corresponding to that keyboard. The intercept device **130** can identify keystrokes received by the patient information system **120**. For example, if the intercept device **130** determines that the keystrokes correspond to regular typing, the intercept device **130** can pass those keystrokes along to whatever application has focus and is listening for them. This would allow, for example, a doctor to type notes in a patient's record. Alternatively, if the intercept device **130** determines that the keystrokes correspond to a unique identifier, then the intercept device **130** can capture the keystroke sequence. In some cases, the intercept device **130** can prevent the keystrokes events from passing to another application running on the patient information system **120**. The intercept device **130** can look up the unique identifier, for example by communicating the unique identifier to the tracking database **142** of the tracking system **140**.

(50) In some cases, the intercept device **130** or the tracking system **140** can identify or determine a generic identifier (for example, a GTIN including the NDC) associated with the unique identifier, and can communicate a sequence of keystrokes to the focus application (for example, the EMR) on the patient information system **120** imitating those keystrokes. In some cases, rather than, or in addition to, identifying a generic identifier, the intercept device **130** or the tracking system **140** can determine types of data that the patient information system **120** is capable of accepting, and can provide the patient information system **120** with pharmacy item data corresponding to those types of data.

(51) In some cases, the intercept device **130** and/or tracking system **140** can be integrated into, or run on, the scanning device **110**. In some such cases, the intercept device **130** or the tracking system **140** can identify or determine the sequence of keystrokes associated with a unique or generic identifier, and can communicate those keystrokes to the patient information system **120**. The patient information system **120** can interpret the keystrokes as a regular barcode scan, for example.

Example Data Flow Diagram

(52) FIG. 2 illustrates a data flow diagram **200** illustrative of an example process for selectively distributing medication data field values to a patient information system **120**. Although FIG. 2 only shows interactions between the scanning device **110**, patient information system **120**, intercept device **130**, and tracking system **140**, the environment **200** can include fewer or more components as desired, or can be configured differently. For example, additional communications can occur between any of the scanning device **110**, patient information system **120**, intercept device **130**, and tracking system **140**, or additional components can be used for communication.

(53) At (1), the scanning device **110** scans one or more machine-readable codes, labels, or tags and retrieves machine-readable data. As described herein, the machine-readable data can include any type of

data, such as a unique identifier, a generic identifier, a National Drug Code (NDC), a Universal Product Number (UPC), patient data, drug data, or the like. The machine-readable data can correspond to a particular medication or item, a particular type of medication, a particular dose of a medication, a particular patient, a particular healthcare professional, a particular location, or the like. The scanning device **110** can communicate at least some of the machine-readable data after the scan.

(54) At (2), the intercept device **130** receives the machine-readable data. The intercept device **130** can receive the machine-readable data from the scanning device **110** or the patient information system **120**. In some cases, the intercept device **130** intercepts the machine-readable data from communications between the scanning device **110** and the patient information system **120**. The intercept device **130** can receive the machine-readable data from the scanning device **110** via a local bus, such as when the intercept device **130** and the scanning device **110** form part of a single device or via a network, such as when the intercept device **130** and the scanning device **110** are distinct devices.

(55) The intercept device **130** can be configured with computer-executable instructions to parse the machine-readable data, for example, to extract or determine the unique identifier. For example, the unique identifiers can be embedded into machine-readable codes or labels. In addition or alternatively, the machine-readable data can be the unique identifier. In some cases, the intercept device **130** can generate the unique identifier, for example, based on the machine-readable data received from the scanning device **110**. For instance, the intercept device **130** can associate unique identifiers with items based on the machine-readable data.

(56) At (3), the intercept device **130** identifies the categories of data that the patient information system **120** uses to store data. As described herein, the patient information system **120** can include a database that stores multiple categories of data for various pharmacy items. The categories, which can be referred to as medication data fields, can include, but are not limited to, drug name, dosage, hospital, bin number, expiration dates, NDC, UPC, GS1 barcode, location, patient or recipient information, recall information, use-by date, or the like. In some cases, the patient information system **120** stores the same categories of data for each pharmacy item on which it stores data. For example, for each pharmacy item, the patient information system **120** can store one or more values (generally referred to as medication data field values) for each category of data that it stores. In some cases, the patient information system **120** stores the different categories of data for different pharmacy items. In some such cases, the intercept device **130** can identify the medication data fields that the patient information system **120** is uses to store medication data field values for the pharmacy item that corresponds to the unique identifier.

(57) The intercept device **130** can identify the categories of data that the patient information system **120** is uses to store data using any of various techniques. For example, the intercept device **130** can query the patient information system **120** to request an indication of the categories. In some instances, the intercept device **130** can query the patient information system **120** by providing the patient information system **120** with the unique identifier. In response to the query, or in some cases without receiving the query, the patient information system **120** can communicate an indication of the categories it uses. As another example, to identify the categories the intercept device **130** can consult a lookup table. For example, the intercept device **130** may have previously communicated with the patient information system **120**, identified the categories, and stored an indication of the categories in a lookup table or other storage.

(58) At (4), the intercept device **130** communicates a request for medication data field values that correspond to the categories identified at step (3) to the tracking system **140**. The request can identify the pharmacy item that corresponds to the unique identifier. For example, the request can include the unique identifier or other indication of the pharmacy item. In some cases, the request can indicate the categories identified at step (3). In some cases, the request may not indicate the categories. For example, the intercept device **130** can be configured to filter out or ignore any medication data field values received from the tracking system **140** that do not correspond to the categories identified at step (3).

(59) The intercept device **130** can communicate the request to the tracking system **140** via a local bus, such as when the intercept device **130** and the tracking system **140** form part of a single device or via a network, such as when the intercept device **130** and the tracking system **140** are distinct devices.

(60) At (5), the tracking system **140** communicates medication data field values to the intercept device **130**. The tracking system **140** can receive the request from the intercept device **130** and can identify the medication data field values based on the request. For example, the tracking system **140** can identify those

medication data field values that correspond to the unique identifier and correspond to the medication data field used by the patient information system **120**. In some cases, the tracking database **142** can use the unique identifier to identify a generic identifier that corresponds to the unique identifier. The generic identifier can be used by the patient information system **120** to obtain additional data from another database, such as an NDC database.

(61) At (6), the intercept device **130** communicates at least some of the medication data field values it received at step (5) to the patient information system **120**. In some cases, the intercept device **130** communicates only medication data field values that it has determined are used by the patient information system **120**.

(62) In some cases, the intercept device **130** communicates at least one medication data field value that corresponds to a medication data field that the patient information system **120** does not use. In some such cases, the patient information system **120** may discard or ignore those medication data field values as superfluous. In this way, the patient information system **120** can filter or select the medication data field values that correspond to medication data fields that the patient information system **120** uses. In some cases, the patient information system **120** may expand its database to include one or more additional medication data fields so that it can store the those additional medication data field values.

(63) As a non-limiting example, if the intercept device **130** determines that the patient information system **120** includes medication data fields for UPC and drug name, but does not include a medication data field for expiration date, then the intercept device **130** may selectively request medication data field values for UPC and drug name from the remotely located database, without making a request for medication data field values corresponding to expiration date.

(64) In certain embodiments, any interactions (1)-(6) may be omitted, occur concurrently, or occur in a different order, as desired. For example, the intercept device **130** can identify the categories of data accepted by the patient information system **120** prior to, or concurrently with, any of interactions (1) or (2). Furthermore, it will be understood that fewer, more, or different interactions may occur as part of the data flow diagram **200**. For example, in some cases, the tracking system **140** can update or create database entries associated with the pharmacy item in the tracking database **142** to indicate that the pharmacy item has reached a particular location, has been placed in the custody of a particular person, or has entered or completed a particular processing step. For instance, the tracking system **140** can use the request to update or create database entry indicating that a dose is being administered to the patient.

(65) As another example, in some cases, the tracking system **140** can verify the particular item based on the request. For example, the unique identifier can be used to ensure that a particular pharmacy item or dose of a pharmacy item is at a correct location within a healthcare facility (for example, at a correct nursing station in route to a patient or in the proper patient's hospital room), that custody of the pharmacy item or dose of a pharmacy item has been given to a proper healthcare professional, (for example, a nurse assigned to the particular patient for whom the pharmacy item has been prepared), that any timing requirements associated with the pharmacy item or dose of a pharmacy item are met, (for example, ensure that medications are timely administered to the patient), that the pharmacy item or dose of a pharmacy item is administered to the correct patient, or that the pharmacy item or dose of a pharmacy item is safe for administration to the patient, (for example, by checking expiration dates, beyond use dates, and blacklist/recall statuses). Other verifications can also be performed.

(66) As another example, the intercept device **130** or the tracking system **140** one or more safety and inventory checks. For instance, the intercept device **130** or the tracking system **140** can verify whether the medication corresponding to the pharmacy item **302** may be expired, have exceeded its beyond use dating, or be flagged as a recalled/blacklisted item. In some cases, if the intercept device **130** or the tracking system **140** determines one or more safety and inventory check errors, the intercept device **130** or the tracking system **140** can cause an alert, for example to the patient or caregiver, or prevent an action, for example by not passing the keystrokes on to another application (e.g., an EMR) of the patient information system **120**.

(67) Flow Diagram

(68) FIG. **3** is a flow diagram illustrating an embodiment of a routine **300** for selectively communicating medication data field values based on the medication data fields that a patient information system **120** uses to store data. In this way, the intercept device **130** can provide varying levels of detail about a

pharmacy interface to the patient information system based on the capabilities or configuration of the patient information system **120**. One skilled in the relevant art will appreciate that the elements outlined for routine **300** can be implemented by one or more computing devices that are associated with the environment **100**, including, but not limited to, the scanning device **110**, the patient information system **120**, the intercept device **130**, the tracking system **140**, or any combination thereof. For simplicity, routine **300** has been associated as being generally performed by the intercept device **130**. However, the following illustrative embodiments should not be construed as limiting.

(69) At block **305**, the intercept device **130** receives an identifier. The identifier can be part of machine-readable data retrieved by a scanning device **110** after it scans one or more machine-readable codes, labels, or tags. In some cases, the intercept device **130** receives the identifier, or the machine-readable data including the identifier, as part of communications from the scanning device **110** to the intercept device **130**. In some cases, the intercept device **130** receives the identifier by intercepting communications between the scanning device **110** and the patient information system **120**. As described, the can correspond to, among other things, a particular drug, a particular type of drug, a particular medicinal container including a particular drug, or a particular medical container including a particular type of drug. Further, the identifier can uniquely identify a medicinal container from other medicinal containers that include the same drug or same type of drug or can uniquely identify a medicinal container from all other medicinal containers.

(70) At block **310**, the intercept device **130** determines one or more medication data fields that a patient information system **120** uses to store data. As described herein, the patient information system **120** can include a database, such as a database for electronic medical records (EMRs), and the database can use a particular set of medication data fields for storing medication data field values. In some cases, the patient information system **120** uses the same medication data fields for each item. In some cases, the patient information system **120** may use different same medication data fields for different items. In some such cases, the intercept device **130** can determine the one or more medication data fields that the patient information system uses to store medication data field values for the particular item that corresponds to the unique identifier.

(71) The intercept device **130** may determine the one or more medication data fields that a patient information system **120** uses to store data using any of various techniques. For example, the intercept device **130** can query the patient information system **120** to request an indication of the one or more medication data fields. In some instances, the intercept device **130** can query the patient information system **120** by providing the patient information system **120** with the unique identifier. In response, the patient information system **120** can communicate an indication of the medication data fields it uses to store data, such as the more medication data fields it uses to store medication data field values for the particular item that corresponds to the unique identifier.

(72) As another example, the intercept device **130** can consult a lookup table to determine the medication data fields. For example, the intercept device **130** may have previously communicated with the patient information system **120** and may already know which medication data fields the patient information system **120** uses to store data and/or is capable of receive.

(73) The intercept device **130** can identify the medication data fields that the patient information system **120** uses, or is configured use, to store data at any time, such as before, after, or while concurrently receiving the machine-readable data from the scanning device **110**. For example, the intercept device **130** can be configured to determine the medication data fields or medication data field values accepted by the patient information system **120** as part of a set up procedure and/or prior to the scanning device **110** performing a scan.

(74) At block **315**, the intercept device **130** queries a remotely located database for one or more medication data field values associated with the one or more medication data fields determined at block **310**. In some cases, the data is stored locally.

(75) As a non-limiting example, the remotely located database may include medication data fields for UPC, drug name, and expiration date. If the intercept device **130** determines that the patient information system **120** includes medication data fields for UPC and drug name, but does not include a medication data field for expiration date, then the intercept device **130** may selectively request medication data field values for UPC and drug name from the remotely located database, without making a request for

medication data field values corresponding to expiration date.

(76) In some cases, the intercept device **130** may request and/or the remotely located database may communicate medication data field values that do not correspond to medication data fields that the patient information system **120** uses or is configured to use. In some such cases, the intercept device **130** may discard or ignore those medication data field values as superfluous. In this way, the intercept device **130** can filter the medication data field values that correspond to medication data fields that the patient information system **120**.

(77) At block **320**, the intercept device **130** receives one or more medication data field values corresponding to the query. In some cases, the intercept device **130** may receive medication data field values that do not correspond to medication data fields that the patient information system **120** uses or is configured to use. In some such cases, the intercept device **130** may discard or ignore those medication data field values as superfluous. In this way, the intercept device **130** can filter or select the medication data field values that correspond to medication data fields that the patient information system **120** uses.

(78) At block **325**, the intercept device **130** communicates the one or more medication data field values to the patient information system **120**. In some cases, the intercept device **130** may communicate medication data field values that do not correspond to medication data fields that the patient information system **120** uses or is configured to use. In some such cases, the patient information system **120** may discard or ignore those medication data field values as superfluous. In this way, the patient information system **120** can filter or select the medication data field values that correspond to medication data fields that the patient information system **120** uses.

(79) It will be understood that the various blocks described herein can be implemented in a variety of orders, and that the intercept device **130** can implement one or more of the blocks concurrently and/or change the order, as desired. For example, the intercept device **130** determine that medication data fields that the patient information system uses prior to, or concurrently with, receiving the identifier. Furthermore, it will be understood that fewer, more, or different blocks can be used as part of the routine **800**. For example, the routine **1600** can include blocks for querying the patient information system **120** to determine which medicate data fields it uses.

Example Embodiments

(80) Various examples of methods and systems for selectively communicating medication data field values to a patient information system can be found in the following clauses:

(81) Clause 1. A method comprising: receiving, from a scanning device, a first identifier corresponding to a computer-readable code associated with a medicinal container including a particular medication, wherein the first identifier uniquely identifies the medicinal container from other medicinal containers; determining one or more medication data fields that a patient information system uses to store one or more medication data field values for the particular medication, wherein the patient information system is communicatively coupled with the scanning device; querying, based at least in part on the first identifier, a remotely located database for the one or more medication data field values for the particular medication; receiving the one or more medication data field values for the particular medication; and communicating the one or more medication data field values to the patient information system.

(82) Clause 2. The method of clause 1, wherein the first identifier uniquely identifies the medicinal container from other medicinal containers that include the same type of drug.

(83) Clause 3. The method of any of the previous clauses, wherein the first identifier uniquely identifies the medicinal container from other medicinal containers that include the same drug.

(84) Clause 4. The method of any of the previous clauses, wherein the scanning device comprises an RFID reader.

(85) Clause 5. The method of any of the previous clauses, wherein the scanning device comprises a barcode scanner.

(86) Clause 6. The method of clause 5, wherein the computer-readable code is a one-dimensional barcode or a two-dimensional barcode.

(87) Clause 7. The method of any of the previous clauses, wherein said querying the remotely located database comprises transmitting the first identifier to a remotely-located tracking system.

(88) Clause 8. The method of clause 7, wherein the remotely-located tracking system is configured to: identify the one or more medication data field values for the particular medication based at least in part on

the first identifier, and communicate the one or more medication data field values.

(89) Clause 9. The method of any of the previous clauses, wherein said determining one or more medication data fields comprises querying the patient information system database for the one or more medication data fields.

(90) Clause 10. The method of any of Clauses 7-9, wherein the remotely-located tracking system is configured to update location information associated with the particular medication.

(91) Clause 11. The method of any of the previous clauses, wherein the medication data field values comprise an NDC number.

(92) Clause 12. The method of any of the previous clauses, wherein the medication data field values comprise a name of the particular drug.

(93) Clause 13. The method of any of the previous clauses, wherein the medication data field values comprise recall information.

(94) Clause 14. The method of any of the previous clauses, wherein the medication data field values comprise at least one of a use-by date or an expiration date.

(95) Clause 15. The method of any of the previous clauses, wherein the patient information system is an electronic medical records database.

(96) Clause 16. A system comprising: an intercept device communicatively coupled along a communication pathway between a scanning device and a patient information system, the intercept device comprising a processor configured to: receive, from the scanning device, a first identifier corresponding to a computer-readable code associated with a medicinal container including a particular medication, wherein the first identifier uniquely identifies the medicinal container from other medicinal containers; determine one or more medication data fields that the patient information system uses to store one or more medication data field values for the particular medication, wherein the patient information system is communicatively coupled with the scanning device; query, based at least in part on the first identifier, a remotely located database for the one or more medication data field values for the particular medication; receive the one or more medication data field values for the particular medication; and communicate the one or more medication data field values to the patient information system.

(97) Clause 17. The system of clause 16, wherein the first identifier uniquely identifies the medicinal container from other medicinal containers that include the same type of drug.

(98) Clause 18. The system of any of clauses 16 or 17, wherein the first identifier uniquely identifies the medicinal container from other medicinal containers that include the same drug.

(99) Clause 19. The system of any of clauses 16 to 18, wherein the scanning device comprises an RFID reader.

(100) Clause 20. The system of any of clauses 16 to 19, wherein the scanning device comprises a barcode scanner.

(101) Clause 21. The system of clause 20, wherein the computer-readable code is a one-dimensional barcode or a two-dimensional barcode.

(102) Clause 22. The system of any of clauses 16 to 21, wherein said querying the remotely located database comprises transmitting the first identifier to a remotely-located tracking system.

(103) Clause 23. The system of clause 22, wherein the remotely-located tracking system is configured to: identify the one or more medication data field values for the particular medication based at least in part on the first identifier, and communicate the one or more medication data field values.

(104) Clause 24. The system of clause 22, wherein to determine the one or more medication data fields, the intercept device is configured to query the patient information system database for the one or more medication data fields.

(105) Clause 25. The system of clause 22, wherein the remotely-located tracking system is configured to update location information associated with the particular medication.

(106) Clause 26. The system of any of clauses 16 to 25, wherein the medication data field values comprise an NDC number.

(107) Clause 27. The system of any of clauses 16 to 26, wherein the medication data field values comprise a name of the particular drug.

(108) Clause 28. The system of any of clauses 16 to 27, wherein the medication data field values comprise recall information.

(109) Clause 29. The system of any of clauses 16 to 28, wherein the medication data field values comprise at least one of a use-by date or an expiration date.

(110) Clause 30. The system of any of clauses 16 to 29, wherein the patient information system is an electronic medical records database.

(111) Clause 31. The system of any of clauses 16 to 30, wherein to receive the first identifier, the intercept device is configured to intercept a communication from the scanning device to the patient information system.

(112) Clause 32. The system of any of clauses 16 to 31, wherein the remotely located database stores medication data field values associated with a plurality of medication data fields, wherein the one or more medication data fields that the patient information system uses to store the one or more medication data field values for the particular medication includes some but not all of the plurality of medication data fields.

ADDITIONAL CONSIDERATIONS

(113) Conjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y and at least one of Z to each be present.

(114) Conditional language, such as, among others, “can,” “could,” “might,” or “can,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

(115) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above detailed description using the singular or plural number can also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, covers all of the following interpretations of the word: any one of the items in the list, all of the items in the list, and any combination of the items in the list. Likewise, the term “and/or” in reference to a list of two or more items, covers all of the following interpretations of the word: any one of the items in the list, all of the items in the list, and any combination of the items in the list.

(116) Depending on the embodiment, certain operations, acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (for example, not all are necessary for the practice of the algorithms). Moreover, in certain embodiments, operations, acts, functions, or events can be performed concurrently, for example, through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially.

(117) Systems and modules described herein can comprise software, firmware, hardware, or any combination(s) of software, firmware, or hardware suitable for the purposes described herein. Software and other modules can reside and execute on servers, workstations, personal computers, computerized tablets, PDAs, and other computing devices suitable for the purposes described herein. Software and other modules can be accessible via local memory, via a network, via a browser, or via other means suitable for the purposes described herein. Data structures described herein can comprise computer files, variables, programming arrays, programming structures, or any electronic information storage schemes or methods, or any combinations thereof, suitable for the purposes described herein. User interface elements described herein can comprise elements from graphical user interfaces, interactive voice response, command line

interfaces, and other suitable interfaces.

(118) Further, the processing of the various components of the illustrated systems can be distributed across multiple machines, networks, and other computing resources. In addition, two or more components of a system can be combined into fewer components. Various components of the illustrated systems can be implemented in one or more virtual machines, rather than in dedicated computer hardware systems and/or computing devices. Likewise, the data storage devices shown can represent physical and/or logical data storage, including, for example, storage area networks or other distributed storage systems. Moreover, the connections between the components shown can represent possible paths of data flow, rather than actual connections between hardware. While some examples of possible connections are shown, any of the subset of the components shown can communicate with any other subset of components in various implementations.

(119) Embodiments are also described above with reference to flow chart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products. Each block of the flow chart illustrations and/or block diagrams, and combinations of blocks in the flow chart illustrations and/or block diagrams, can be implemented by computer program instructions. Such instructions can be provided to a processor of a general purpose computer, special purpose computer, specially-equipped computer (for example, comprising a high-performance database server, a graphics subsystem, etc.) or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor(s) of the computer or other programmable data processing apparatus, create means for implementing the acts specified in the flow chart and/or block diagram block or blocks.

(120) These computer program instructions can also be stored in a non-transitory computer-readable memory that can direct a computer or other programmable data processing apparatus to operate in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the acts specified in the flow chart and/or block diagram block or blocks. The computer program instructions can also be loaded onto a computing device or other programmable data processing apparatus to cause a series of operations to be performed on the computing device or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the acts specified in the flow chart and/or block diagram block or blocks.

(121) Any patents and applications and other references noted above, including any that can be listed in accompanying filing papers, are incorporated herein by reference. Aspects of the disclosure can be modified, if necessary, to employ the systems, functions, and concepts of the various references described above to provide yet further implementations of the disclosure.

(122) These and other changes can be made in light of the above detailed description. While the above description describes certain examples of the disclosure, and describes the best mode contemplated, no matter how detailed the above appears in text, the disclosure can be practiced in many ways. Details of the system can vary considerably in its specific implementation, while still being encompassed by the disclosure disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the disclosure should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the disclosure with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the disclosure to the specific examples disclosed in the specification, unless the above detailed description section explicitly defines such terms. Accordingly, the actual scope of the disclosure encompasses not only the disclosed examples, but also all equivalent ways of practicing or implementing the disclosure under the claims.

Claims

1. A method comprising: receiving, from a scanning device, a first identifier corresponding to a computer-readable code associated with a pharmacy item container including a particular pharmacy item, wherein the first identifier uniquely identifies the pharmacy item container from other pharmacy item containers that include the particular pharmacy item and a same amount of the particular pharmacy item as the pharmacy item container; determining a first set of data fields that a patient information system uses to

store a corresponding first set of data field values for the particular pharmacy item, wherein the patient information system is communicatively coupled with the scanning device; querying, based at least in part on the first identifier, a remotely located database for the first set of data field values for the particular pharmacy item, wherein the remotely located database stores a second set of data fields associated with the pharmacy item container and a corresponding second set of data field values, wherein the first set of data field values is a subset of the second set of data field values; receiving the first set of data field values for the particular pharmacy item; and communicating the first set of data field values to the patient information system.

2. The method of claim 1, wherein the scanning device comprises an RFID reader.

3. The method of claim 1, wherein the scanning device comprises a barcode scanner.

4. The method of claim 3, wherein the computer-readable code is a one-dimensional barcode or a two-dimensional barcode.

5. The method of claim 1, wherein querying the remotely located database comprises transmitting the first identifier to a remotely-located tracking system.

6. The method of claim 5, wherein the remotely-located tracking system is configured to: identify the first set of data field values for the particular pharmacy item based at least in part on the first identifier, and communicate the first set of data field values.

7. The method of claim 5, wherein the remotely-located tracking system is configured to update location information associated with the particular pharmacy item.

8. The method of claim 1, wherein determining the first set of data fields comprises querying the remotely located database for the first set of data fields.

9. The method of claim 1, wherein the first set of data field values or the second set of data field values comprise an NDC number.

10. The method of claim 1, wherein the first set of data field values or the second set of data field values comprise a name of the particular pharmacy item.

11. The method of claim 1, wherein the first set of data field values or the second set of data field values comprise recall information.

12. The method of claim 1, wherein the first set of data field values or the second set of data field values comprise at least one of a use-by date or an expiration date.

13. The method of claim 1, wherein the patient information system is an electronic medical records database.

14. A system comprising: an intercept device communicatively coupled along a communication pathway between a scanning device and a patient information system, the intercept device comprising one or more processors configured to: receive, from the scanning device, a first identifier corresponding to a computer-readable code associated with a pharmacy item container including a particular pharmacy item, wherein the first identifier uniquely identifies the pharmacy item container from other pharmacy item containers that include the particular pharmacy item and a same amount of the particular pharmacy item as the pharmacy item container; determine a first set of data fields that the patient information system uses to store a corresponding first set of data field values for the particular pharmacy item, wherein the patient information system is communicatively coupled with the scanning device; query, based at least in part on the first identifier, a remotely located database for the first set of data field values for the particular pharmacy item, wherein the remotely located database stores a second set of data fields associated with the pharmacy item container and a corresponding second set of field values, wherein the first set of data field values is a subset of the second set of data field values; receive the first set of data field values for the particular pharmacy item; and communicate the first set of data field values to the patient information system.

15. The system of claim 14, wherein the scanning device comprises at least one of an RFID reader or a barcode scanner.

16. The system of claim 14, wherein to determine the first set of data fields, the one or more processors are configured to query the remotely located database for the first set of data fields.

17. The system of claim 14, wherein the first set of data field values or the second set of data field values comprise at least one of an NDC number, a name of the particular pharmacy item, recall information, a use-by date, or an expiration date.

18. The system of claim 14, further comprising a second identifier remote from the first identifier and comprising information associated with a particular patient, wherein the one or more processors are further configured to correlate the first identifier and the second identifier to verify patient administration.

19. The system of claim 14, wherein to query the remotely located database the one or more processors are configured to transmit the first identifier to a remotely-located tracking system; and wherein the remotely-located tracking system is configured to: identify the first set of data field values for the particular pharmacy item based at least in part on the first identifier, and communicate the first set of data field values.

20. Non-transitory computer-readable media comprising computer-executable instructions that, when executed by a computing system for determining event discrepancies between data sets from distributed systems, cause the computing system to: receive, from a scanning device, a first identifier corresponding to a computer-readable code associated with a pharmacy item container including a particular pharmacy item, wherein the first identifier uniquely identifies the pharmacy item container from other pharmacy item containers that include the particular pharmacy item and a same amount of the particular pharmacy item as the pharmacy item container; determine a first set of data fields that a patient information system uses to store a corresponding first set of data field values for the particular pharmacy item, wherein the patient information system is communicatively coupled with the scanning device; query, based at least in part on the first identifier, a remotely located database for the first set of data field values for the particular pharmacy item, wherein the remotely located database stores a second set of data fields associated with the pharmacy item container and a corresponding second set of data field values, wherein the first set of data field values is a subset of the second set of data field values; receive the first set of data field values for the particular pharmacy item; and communicate the first set of data field values to the patient information system.
